

Digital Signal Processing - Lecture 2

Convolution

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| Review

Last time: LTI systems

An LTI system has three defining properties.

Do you remember them?

Last time: LTI systems the three properties

A system that creates an output $y[n]$ for a given input $x[n]$ is LTI if it satisfies the properties of linearity and shift invariance. In these slides, $\rightarrow$ indicates the input-output relationship of the system.

Homogeneity

Scale input and output equally.

$$a x[n] \rightarrow a y[n]$$

Additivity

Add inputs and their outputs.

Given $x_1[n] \rightarrow y_1[n]$ and $x_2[n] \rightarrow y_2[n]$

$$x_1[n] + x_2[n] \rightarrow y_1[n] + y_2[n]$$

Shift invariance

Shift input and output equally. Also referred to as time invariance.

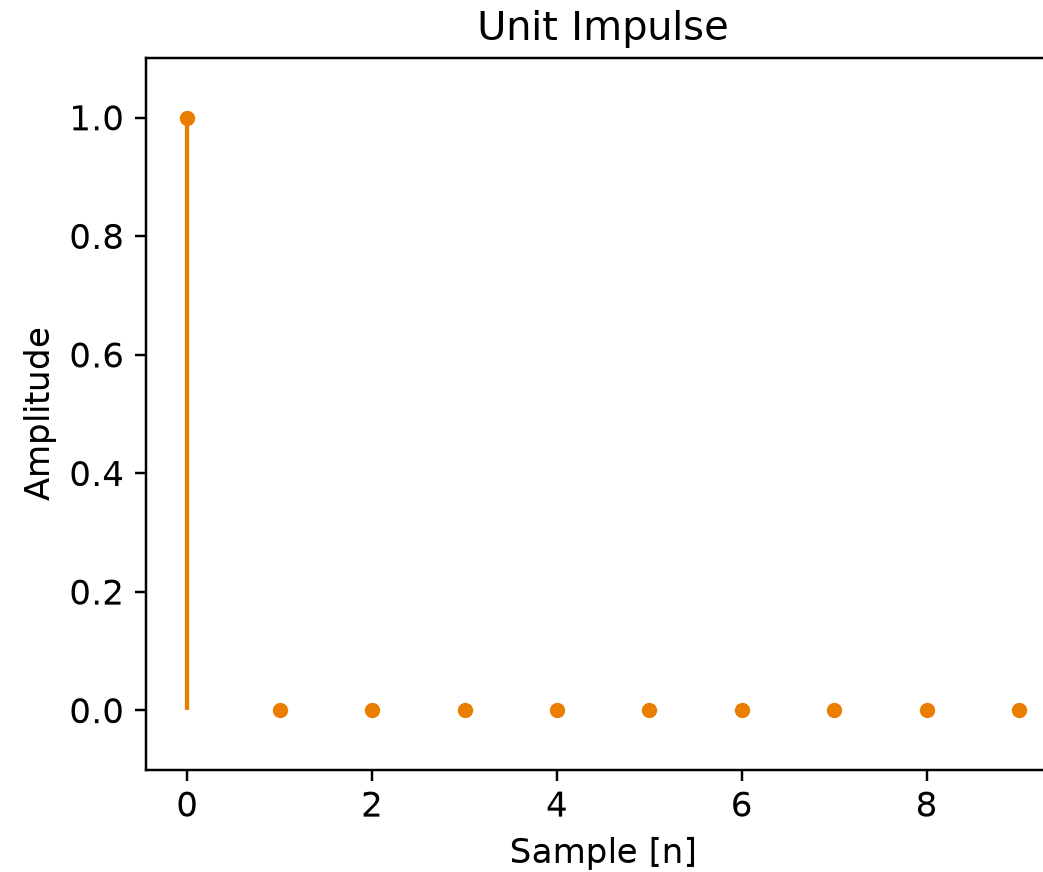
$$x[n - n_0] \rightarrow y[n - n_0]$$

Homogeneity + additivity = linearity

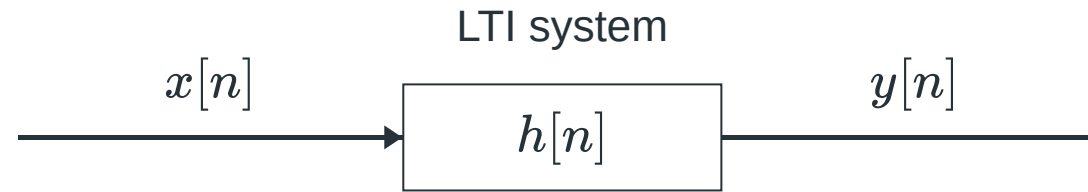
Linearity + shift invariance = LTI

Last time: discrete-time unit impulse

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$



Last time: impulse response



$h[n]$ = response of the LTI system to $\delta[n]$

Questions?

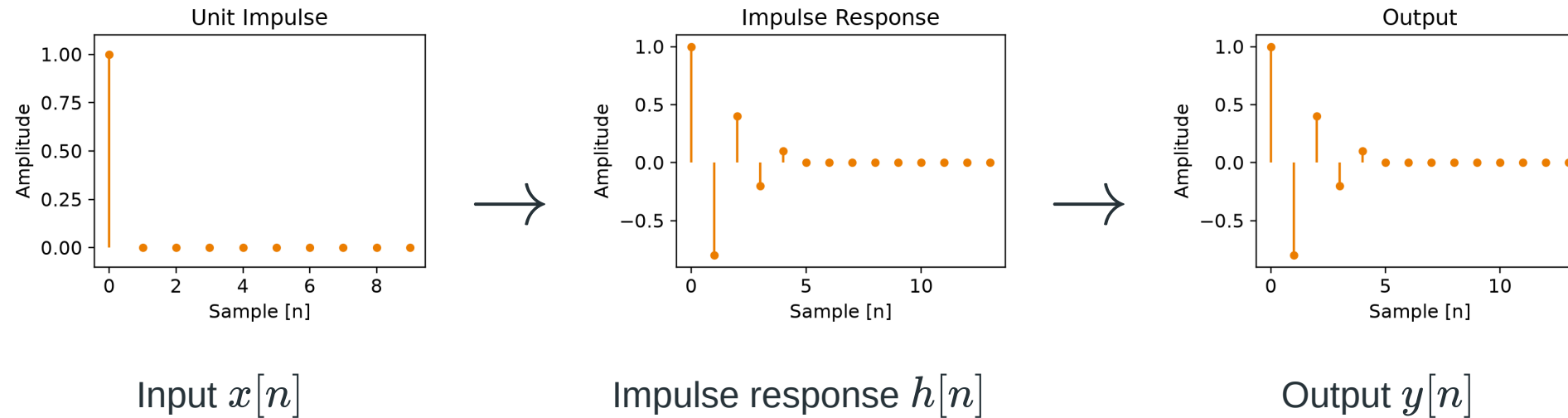
| Why convolution?

Convolution

Convolution in an LTI system computes the output by combining the input signal with shifted and scaled copies of the system's impulse response.

Note, convolution is also used in different contexts. In the grand scheme of things, it is just a relatively fundamental mathematical operation.

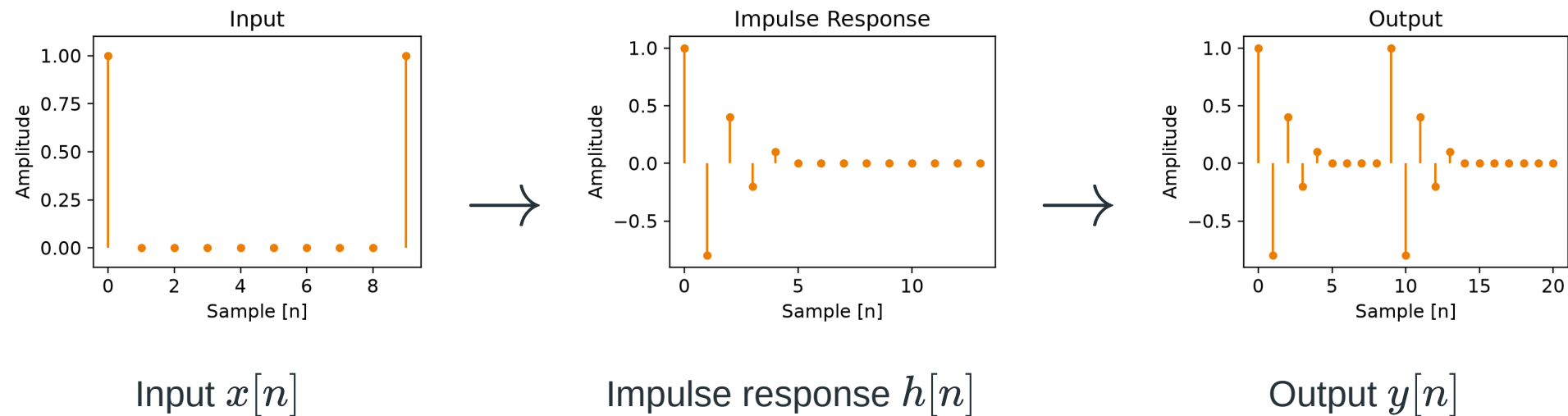
Simple: one unit impulse



$$\delta[n] * h[n] = h[n]$$

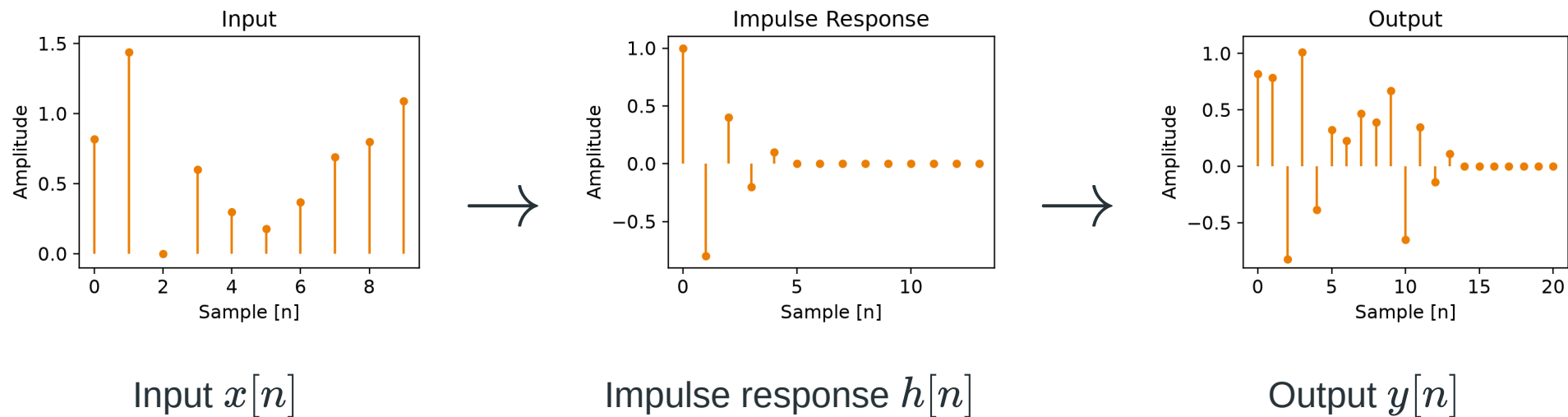
When a unit impulse is applied to an LTI system, the output equals its impulse response, $y[n] = h[n]$.

... Steady ...



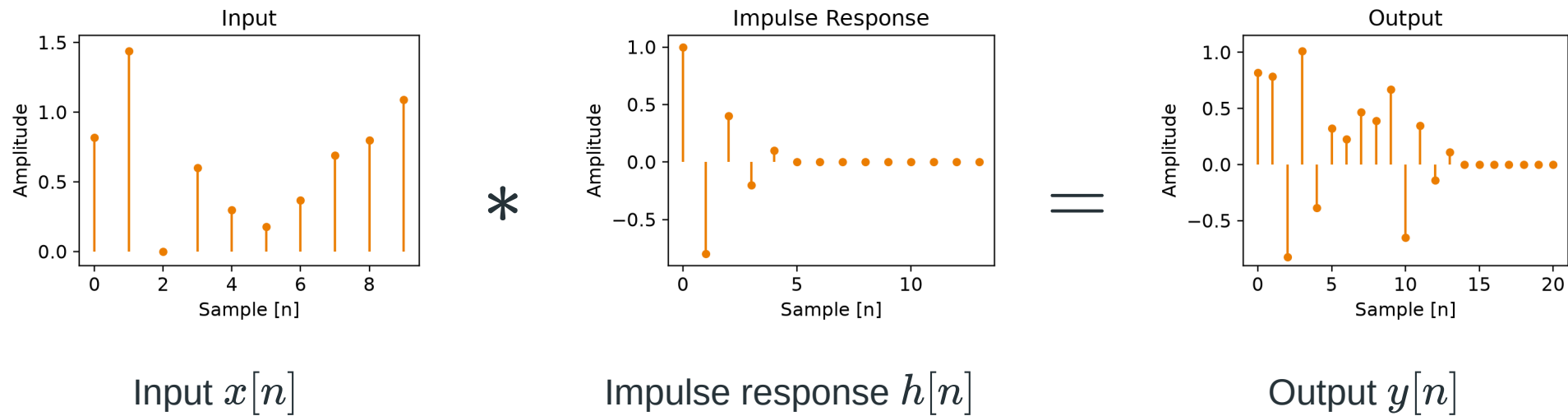
Each shifted impulse produces a shifted, scaled impulse response.

Excuse me?!



How can we combine all the contributions to obtain the output?

Convolution!



$$x[n] * h[n] = y[n]$$

How to convolve?

Discrete time convolution formula

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

| Convolution visualised

Convolution visualisation

$(1, 2, 3) * (4, 5, 6) = (\quad)$

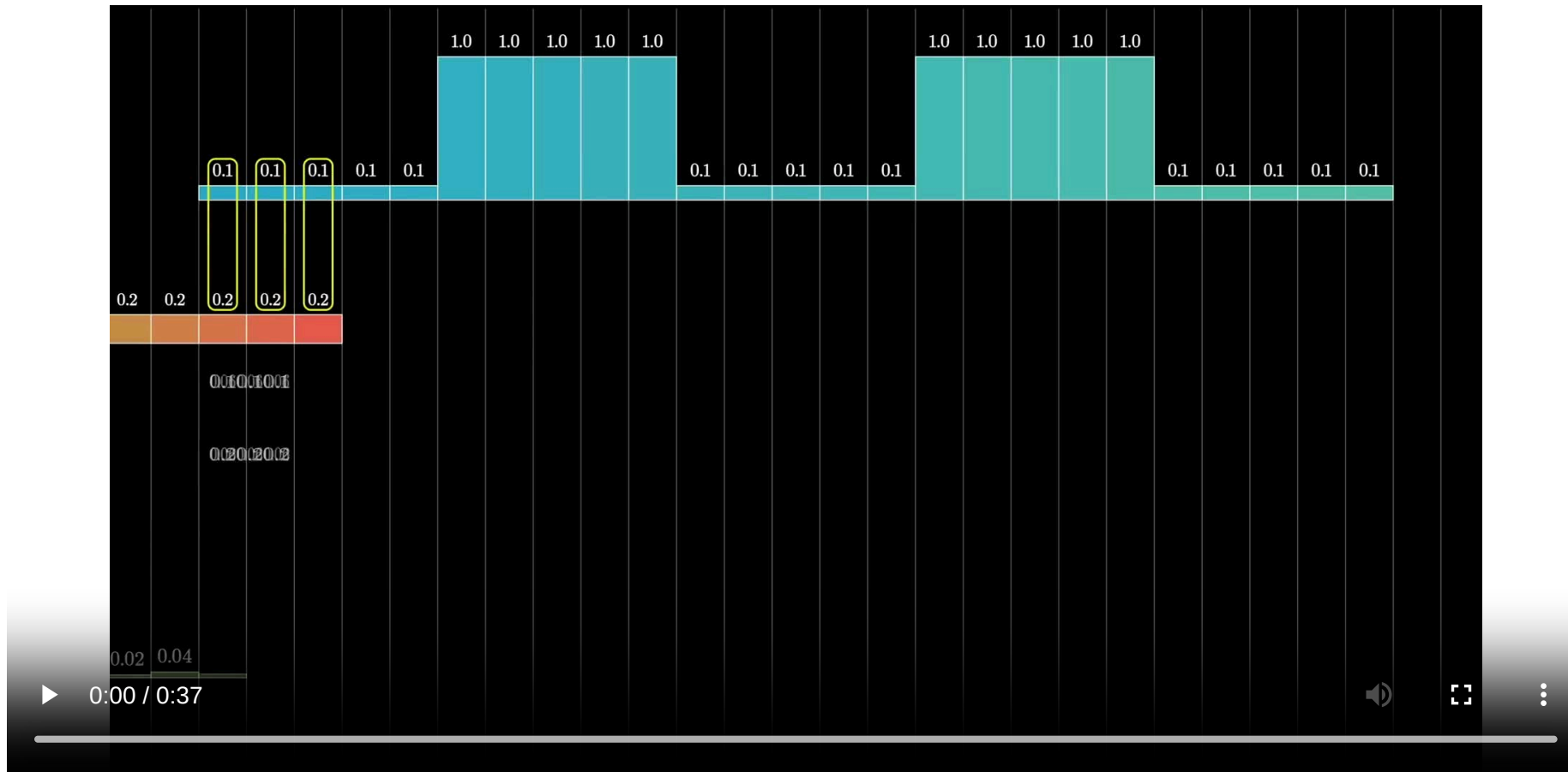
a_0	a_1	a_2
1	2	3

6	5	4
b_2	b_1	b_0

0:00 / 0:41

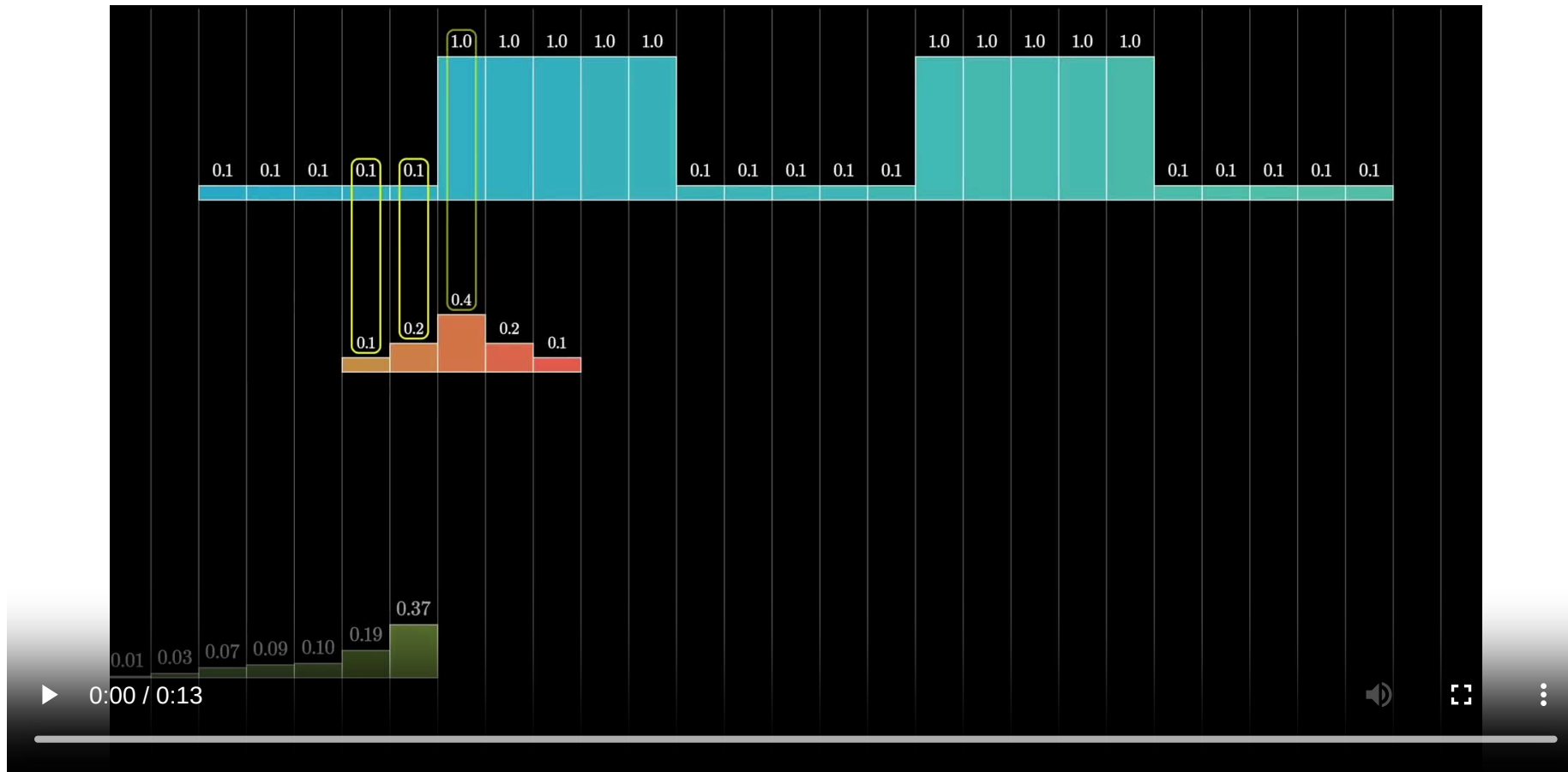
Source: <https://youtu.be/KuXjwB4LzSA>

Convolution moving average



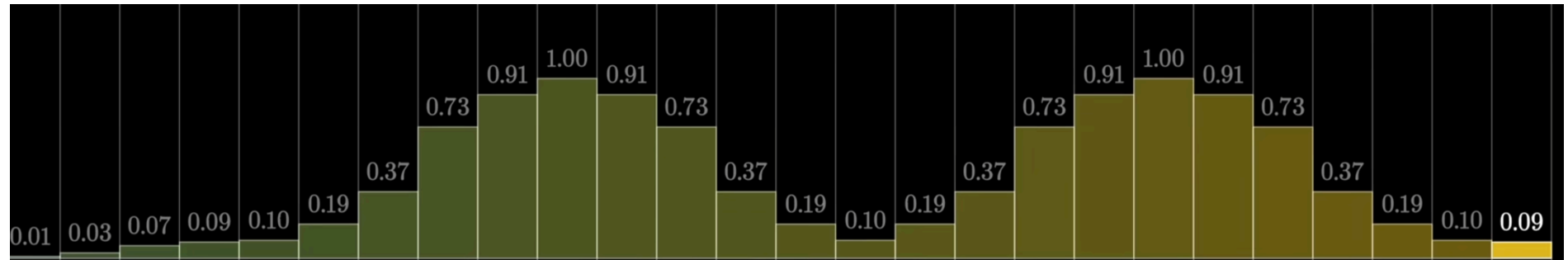
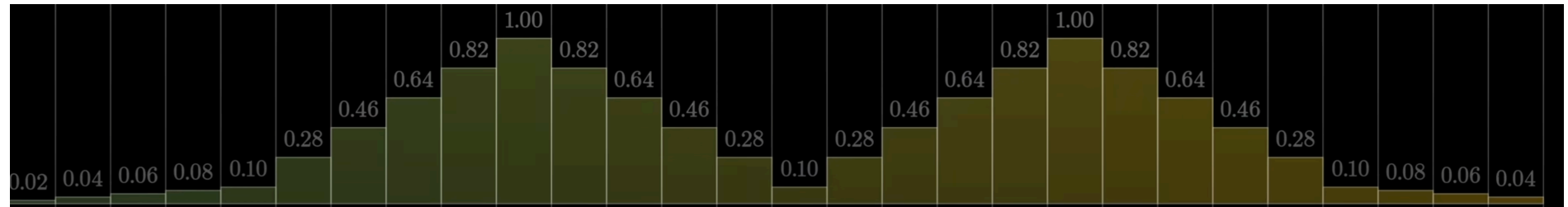
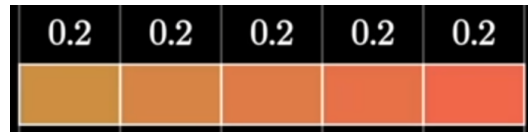
Source: <https://youtu.be/KuXjwB4LzSA>

Convolution weighted average



Source: <https://youtu.be/KuXjwB4LzSA>

Comparing the two averages



Source: <https://youtu.be/KuXjwB4LzSA>

| Convolution calculated

Input and output indices

$$\underbrace{y[n]}_{\text{output sample}} = \sum_{k=-\infty}^{\infty} \underbrace{x[k]}_{\text{input sample}} h[n - k]$$

Choose the output index n . The index k runs through the input samples, pairing each $x[k]$ with $h[n - k]$. For example, for $y[2]$, keep $n = 2$ fixed while k increments.

Fix the output sample

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

Hold n fixed while computing one output value, then move to the next n .

The shifted impulse response

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \underbrace{h[n-k]}_{\text{shifted response}}$$

An input impulse at sample k contributes the response shifted by k samples. Analogous to the rotation shown in the convolution visualisation video.

The summation index

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

- Sum over every input sample k .
- For finite sequences, zero-valued terms contribute nothing.
- Samples outside the given sequences are zero. We only need to multiply and add where both sequences have nonzero values.

Multiply corresponding samples

For $n = 2$, pair $x[k]$ with $h[2 - k]$. Treat values outside the sequences as zero (and products containing zero equal to zero).

$$y[2] = \sum_k x[k]h[2 - k]$$

$$y[2] = \underbrace{x[0]h[2]}_{k=0} + \underbrace{x[1]h[1]}_{k=1} + \underbrace{x[2]h[0]}_{k=2}$$

Each pair's indices add up to 2. First calculate the product for each pair.

Sum the contributions

Each product is one input sample's contribution to $y[2]$.

$$y[2] = \underbrace{x[0]h[2] + x[1]h[1] + x[2]h[0]}_{\text{add the three contributions}}$$

Naturally, adding these products gives one output sample, $y[2]$.

Repeat this calculation for each output index n to obtain the full output sequence.

Example

$$x = [1, 2, 3], \quad h = [4, 5, 6]$$

$$\begin{aligned} y[0] &= x[0]h[0 - 0] &= 1 \cdot 4 &= 4 \\ y[1] &= x[0]h[1 - 0] + x[1]h[1 - 1] &= 1 \cdot 5 + 2 \cdot 4 &= 13 \\ y[2] &= x[0]h[2 - 0] + x[1]h[2 - 1] + x[2]h[2 - 2] &= 1 \cdot 6 + 2 \cdot 5 + 3 \cdot 4 &= 28 \\ y[3] &= x[1]h[3 - 1] + x[2]h[3 - 2] &= 2 \cdot 6 + 3 \cdot 5 &= 27 \\ y[4] &= x[2]h[4 - 2] &= 3 \cdot 6 &= 18 \end{aligned}$$

$$y = [4, 13, 28, 27, 18]$$

All products resulting in zero have been omitted.

Continuous-time convolution

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Integration in continuous time corresponds to summation in discrete time.

| Exercises

Exercises!

`../exercises/DSP-Lecture-02-Exercises.ipynb`